

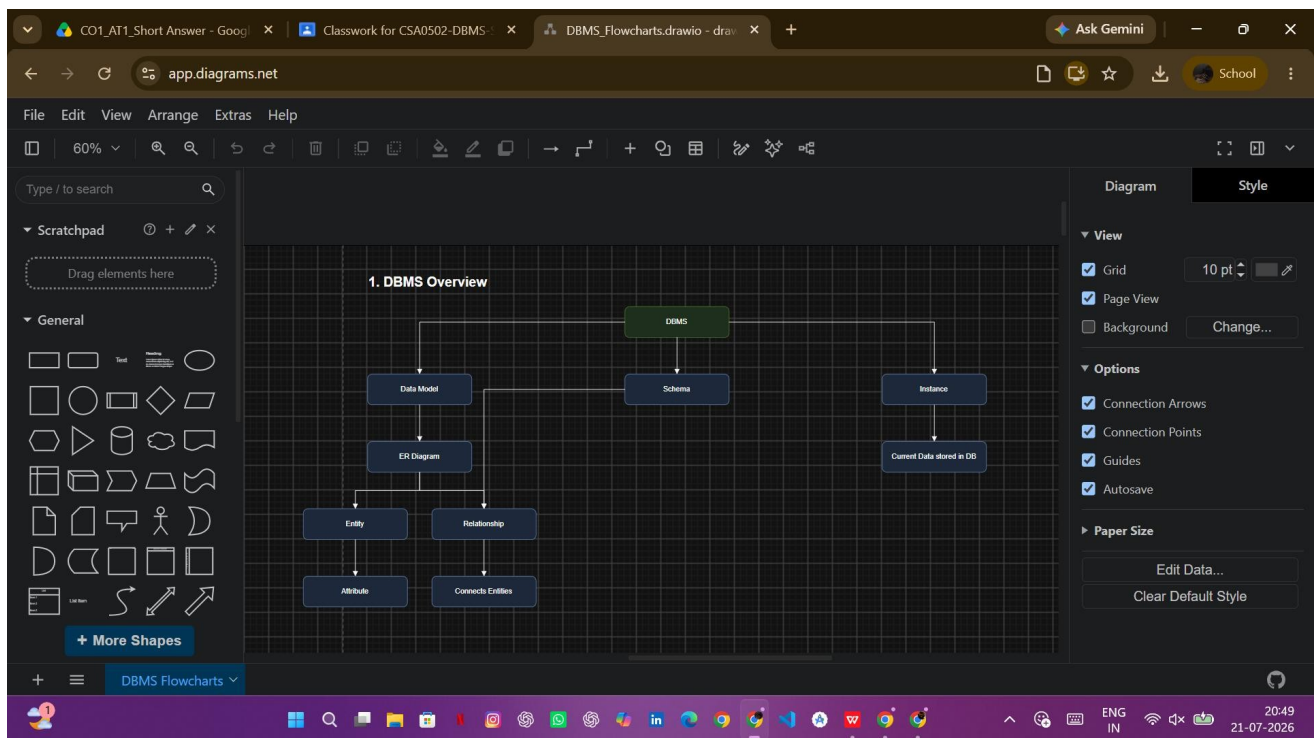
CSA05

DATABASE MANAGEMENT SYSTEM

Assessment Tool 2

**Q1. Construct a concept map linking the following terms:
DBMS, Schema, Instance, Data Model, ER Diagram,
Entity, Attribute, Relationship.**

Concept Map

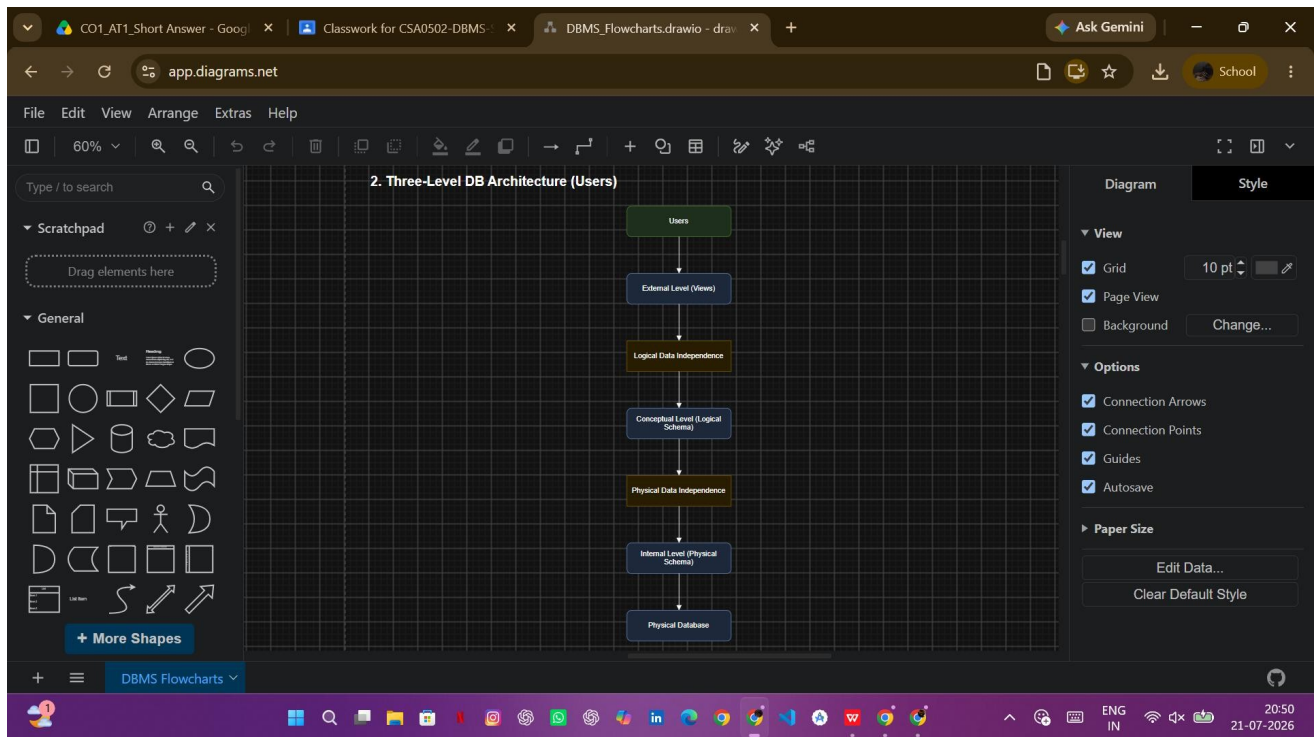


Explanation

- **DBMS** is software used to store, retrieve, and manage data.
- A **Data Model** defines how data is organized.
- The **ER Diagram** represents the conceptual design.
- An **Entity** represents a real-world object.
- **Attributes** describe entities.
- **Relationships** connect entities.
- A **Schema** defines the database structure.
- An **Instance** is the actual data present at a particular time.

Q2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence.

Concept Map



The ANSI/SPARC architecture separates the database into three levels.

External Level

- User-specific views
- Hides unnecessary information

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Conceptual Level

- Complete logical structure
- Defines entities, relationships, and constraints

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Internal Level

- Physical storage details

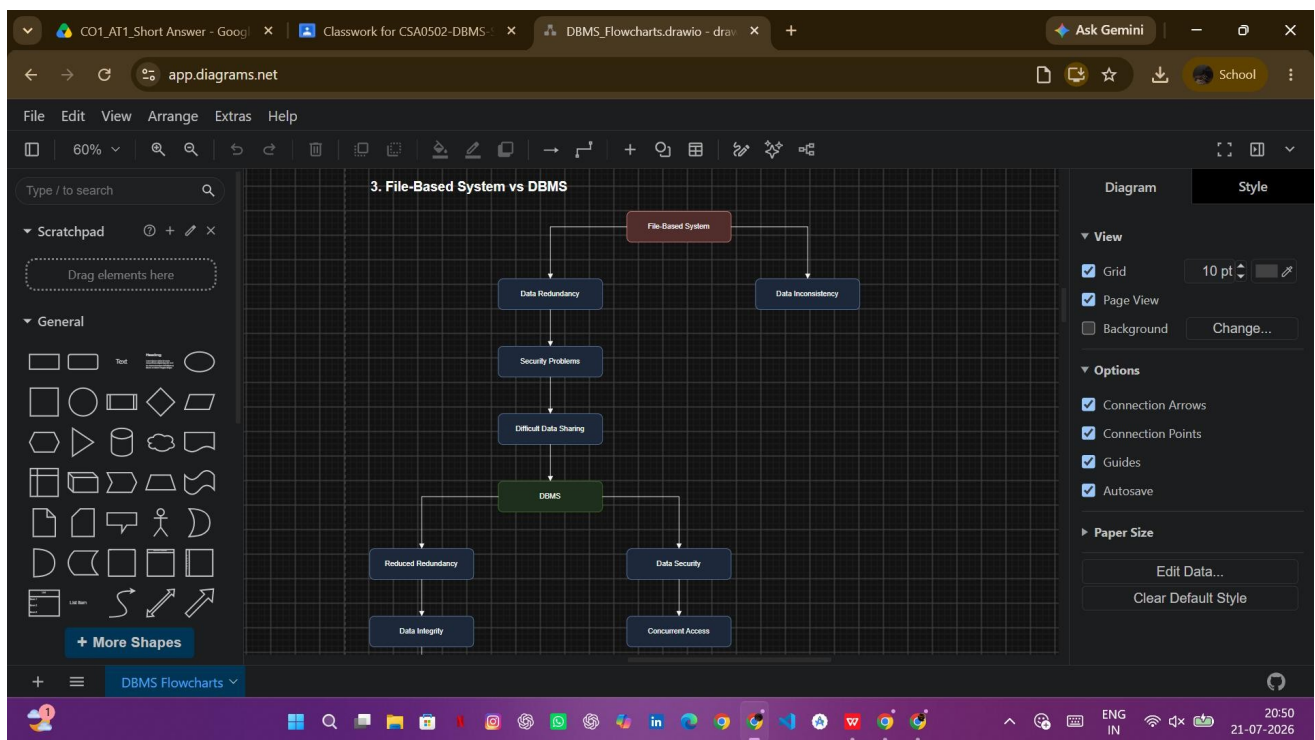
- File organization and indexing

Data Independence

- **Logical Data Independence**
 - Changes in conceptual schema do not affect external views.
- **Physical Data Independence**
 - Changes in storage do not affect conceptual schema.

Q3. Map the relationship between file-based systems and database systems.

Concept Map



Explanation

Problems in File System

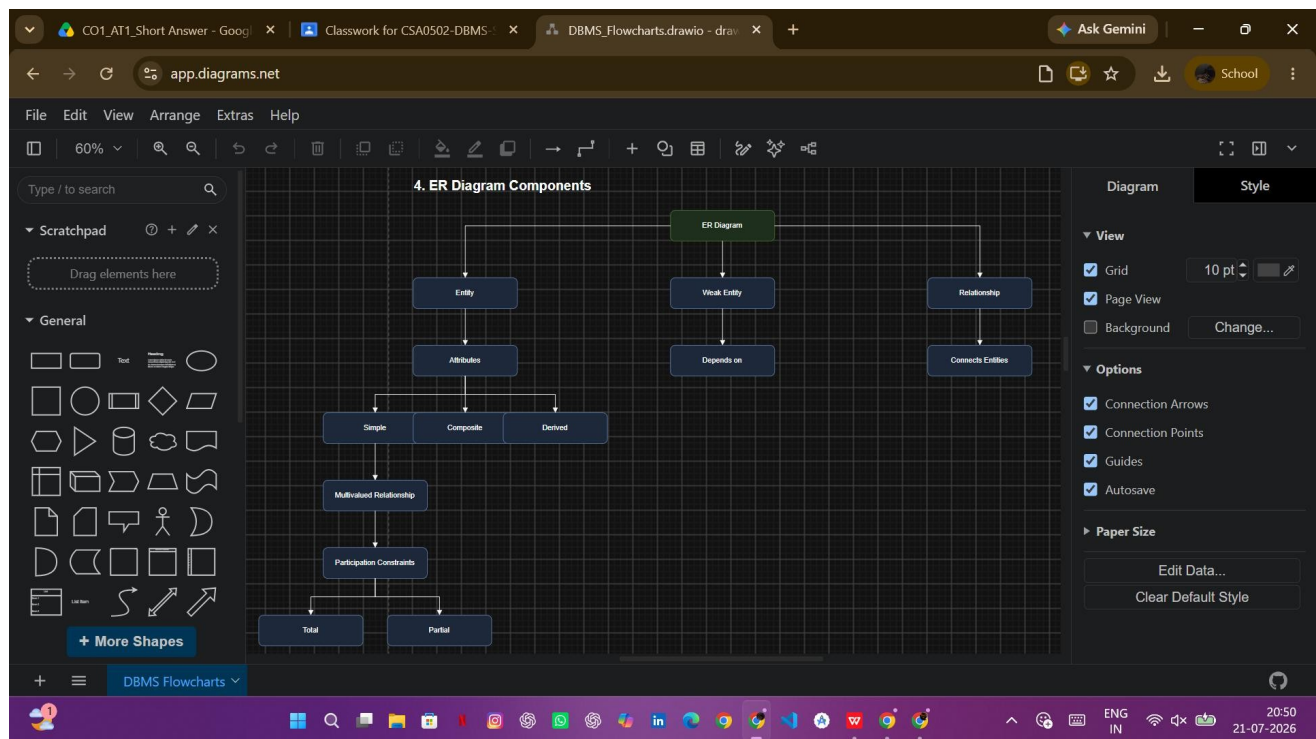
- Data redundancy
- Data inconsistency
- Security issues
- Difficult maintenance
- Poor sharing

DBMS Solutions

- Centralized database
- Reduced redundancy
- Improved security
- Data integrity
- Backup and recovery
- Concurrent access

Q4. Design a concept map for ER diagram components.

Concept Map



Entity Types

- Strong Entity
- Weak Entity

Attribute Types

- Simple
- Composite
- Multivalued
- Derived
- Key Attribute

Relationship Types

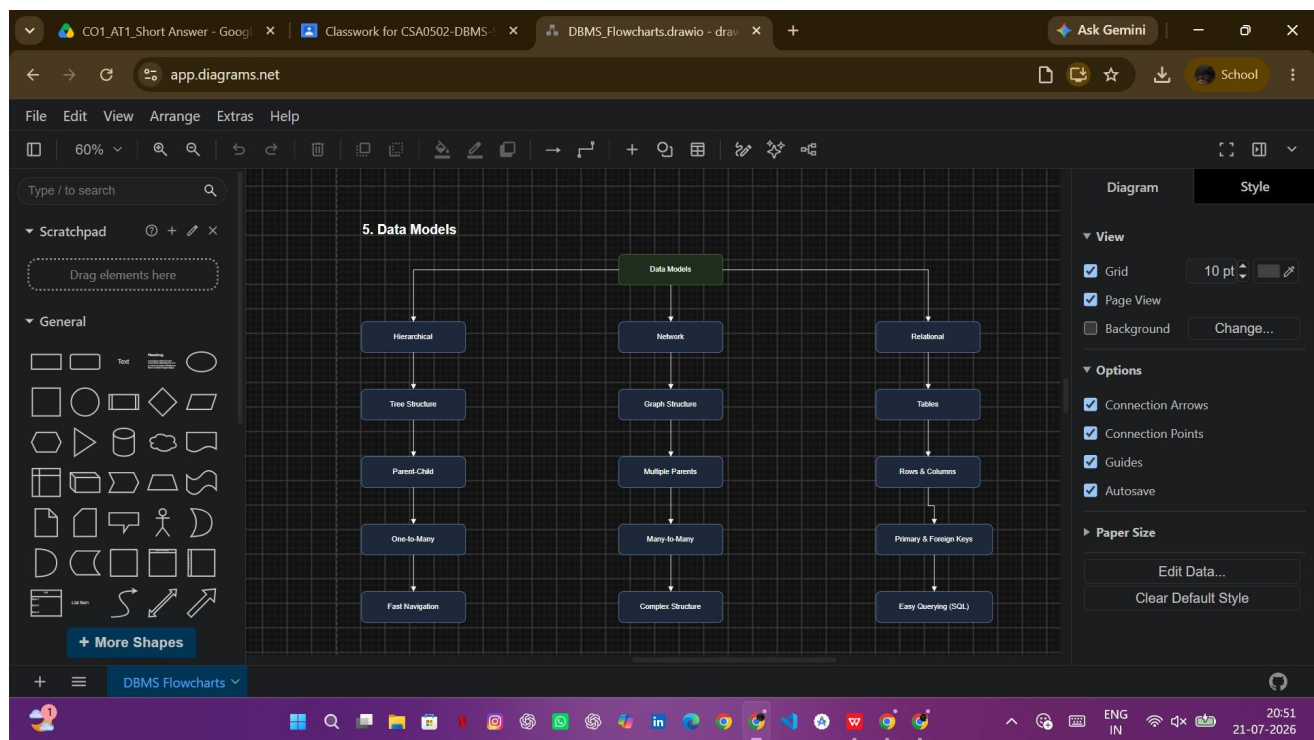
- One-to-One
- One-to-Many
- Many-to-Many

Participation Constraints

- Total Participation
- Partial Participation

Q5. Construct a concept map comparing Hierarchical, Network, and Relational data models.

Concept Map

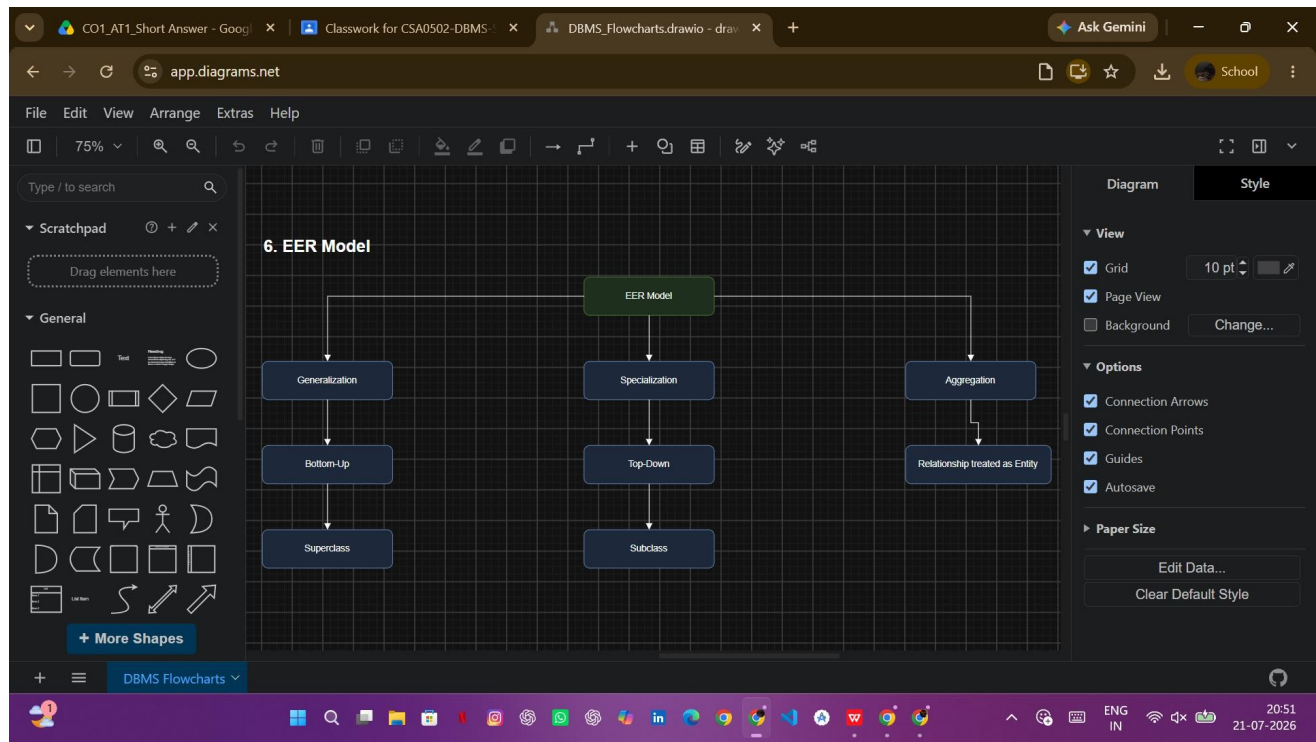


Comparison Table

Hierarchical	Network	Relational
Tree structure	Graph structure	Table structure
One parent	Multiple parents	Relations using keys
Fast access	Flexible	Easy to use
Difficult modifications	Complex	Most widely used

Q6. Develop a concept map for the EER model.

Concept Map



Explanation

The Enhanced ER (EER) model extends the ER model.

Features

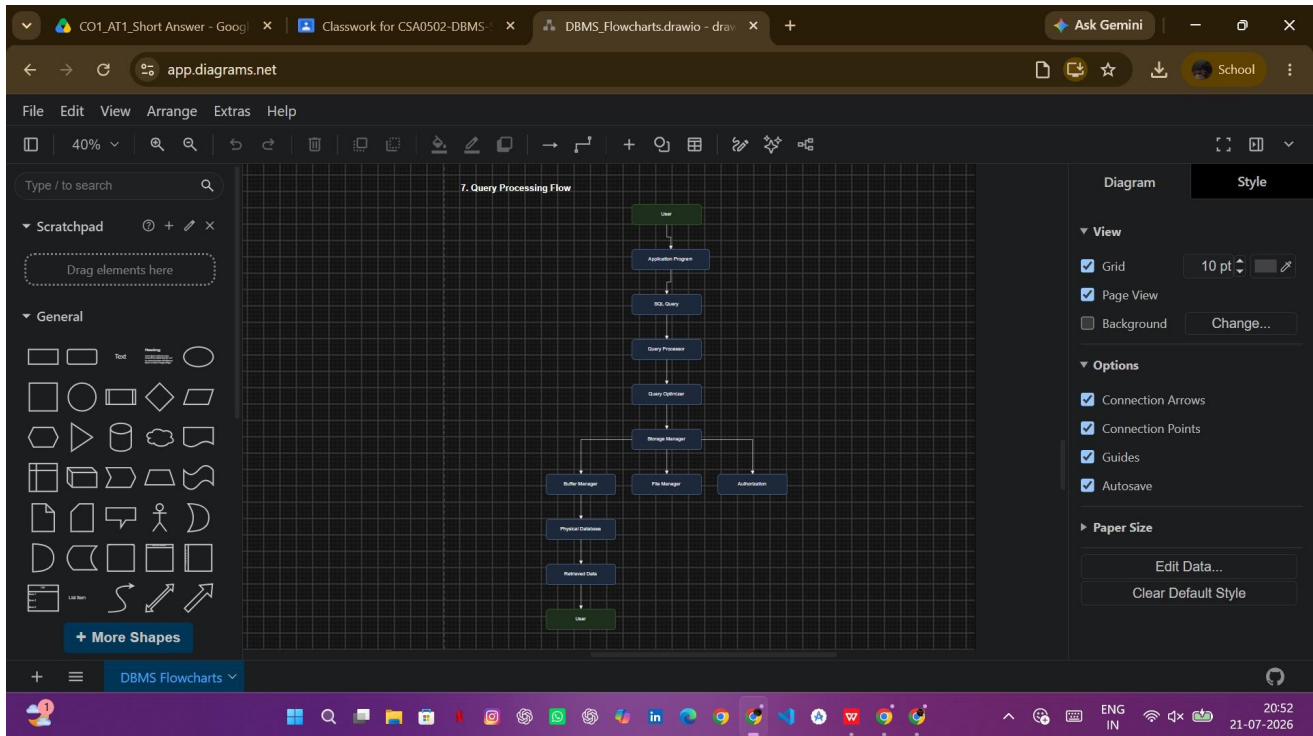
- Generalization
- Specialization
- Aggregation
- Inheritance

Advantages

- Better real-world representation
- Reduced redundancy
- Improved database design

Q7. Create a visual concept map for DBMS software architecture.

Concept Map



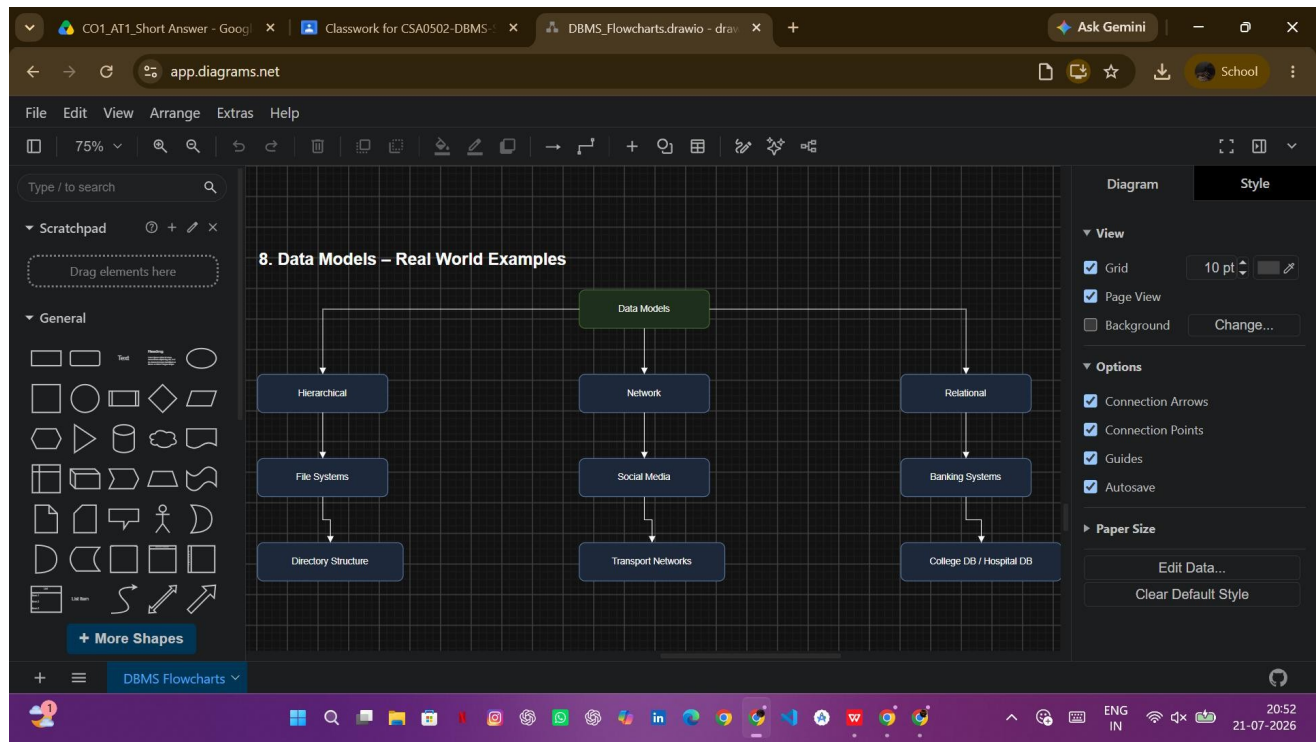
Explanation

Flow

User → Application → SQL Query → Query Processor → Optimizer → Storage Manager → Physical Database → Results

Q8. Map the data models to real-world applications.

Concept Map



Applications

Hierarchical Model

- Windows Directory
- XML Documents

Network Model

- Airline Reservation
- Railway Network
- Social Network

Relational Model

- Banking
- Hospital
- College
- Library
- E-Commerce

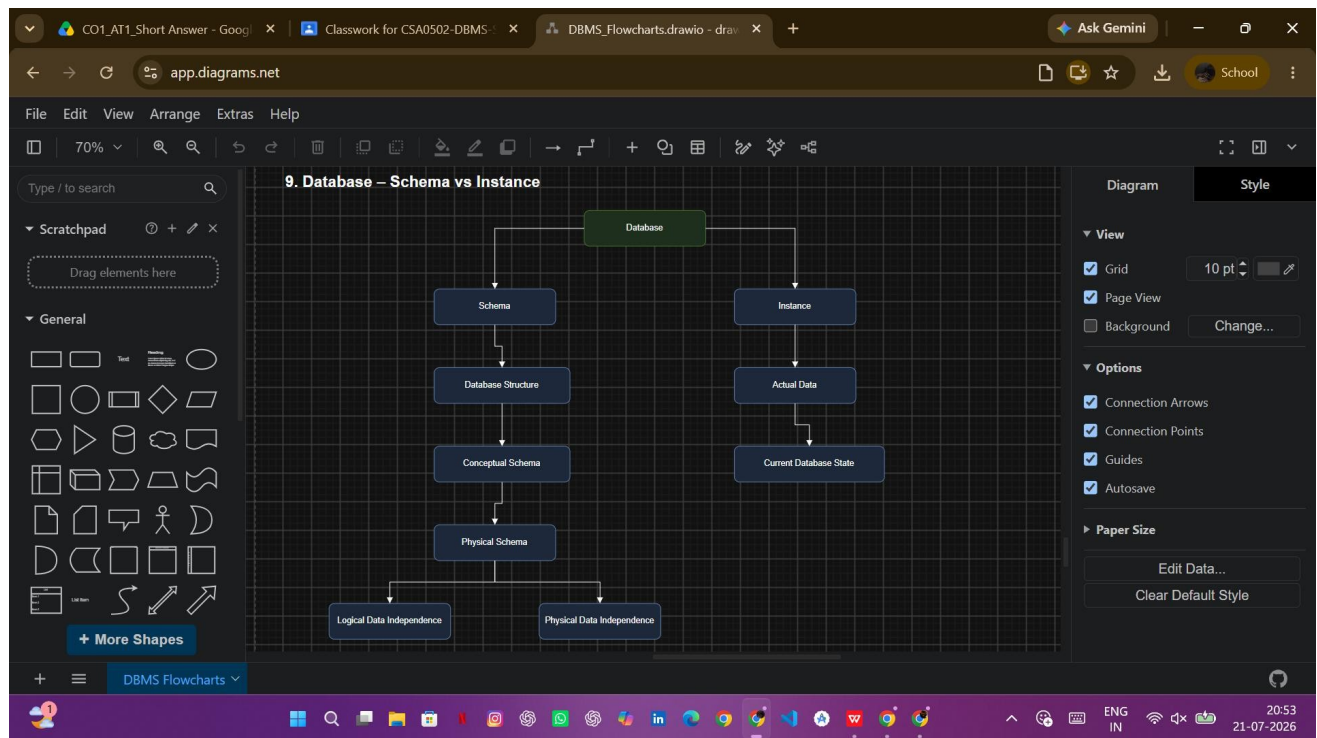
Suitability

- Hierarchical → Best for tree structures.
- Network → Best for many-to-many relationships.

- Relational → Best for business applications.

Q9. Construct a concept map for schemas and instances.

Concept Map



Explanation

Schema

- Database blueprint
- Changes rarely

Instance

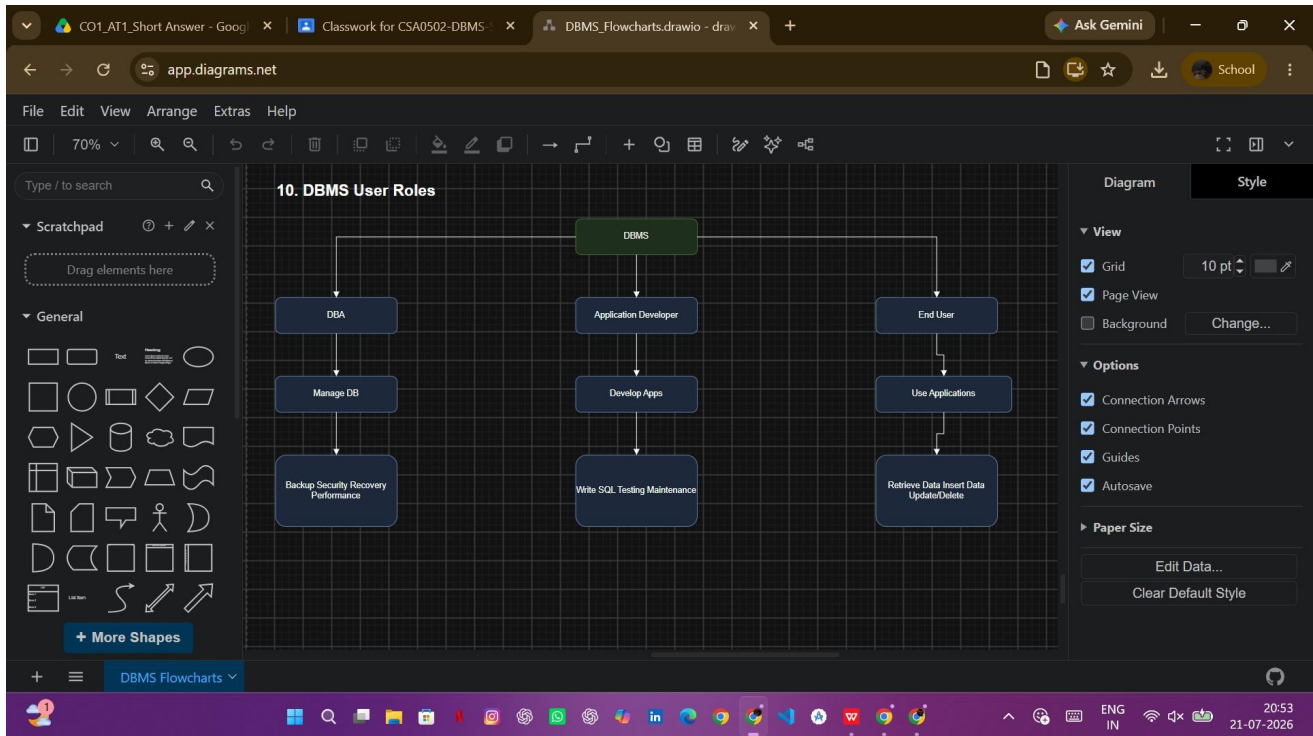
- Current database contents
- Changes frequently

Data Independence

- Logical
- Physical

Q10. Design a concept map for the roles and responsibilities in a DBMS environment.

Concept Map



Performance

Responsibilities

Database Administrator (DBA)

- Database installation
- Security management
- Backup and recovery
- Performance tuning
- User management

Application Developer

- Designs applications
- Writes SQL queries
- Tests applications
- Maintains software

End User

- Accesses data
- Inserts records
- Updates records

- Generates reports

Conclusion

Database Management Systems provide an efficient and secure way to manage data. Concept maps help visualize the relationships among database concepts such as schemas, data models, ER/EER diagrams, software architecture, and user roles. Understanding these relationships improves database design and implementation skills, making DBMS more effective in real-world applications.